Benchmarking Classical Grid-Based Planners in ROS 2

Name: Mohamed Eljahmi
Course: RBE 550 – Motion Planning
Professor: Dan Flickinger
Date: October 7, 2025

Motivation

- **Motion planning algorithms trade off between speed, optimality, and memory.**
- **Benchmarking these planners fairly is difficult without consistent environments.**
- **This project creates an automated, reproducible ROS 2 benchmarking framework.**

Objectives

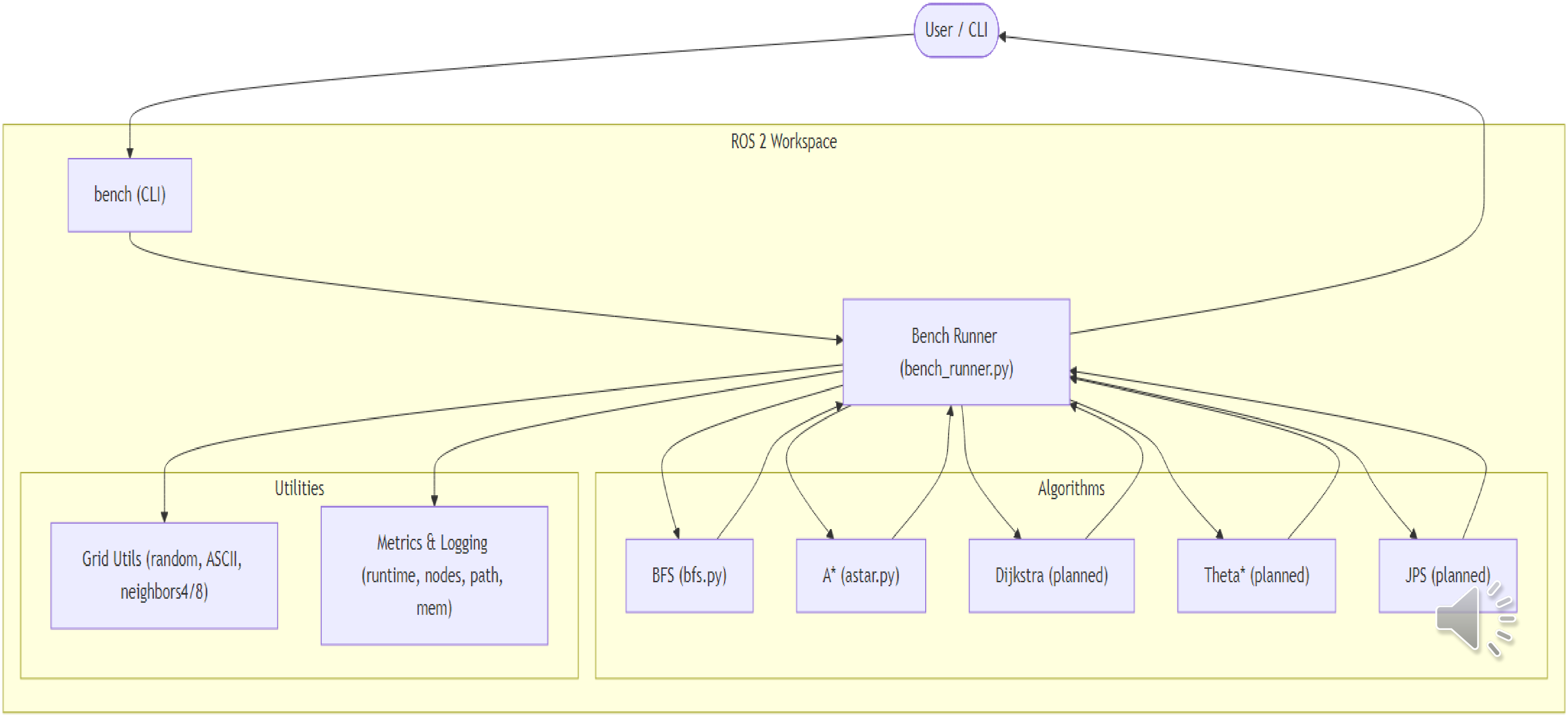
- Implement multiple search algorithms:
  BFS, **Dijkstra**, **Greedy Best-First**, **A***.

- Generate both random and structured (maze) environments.

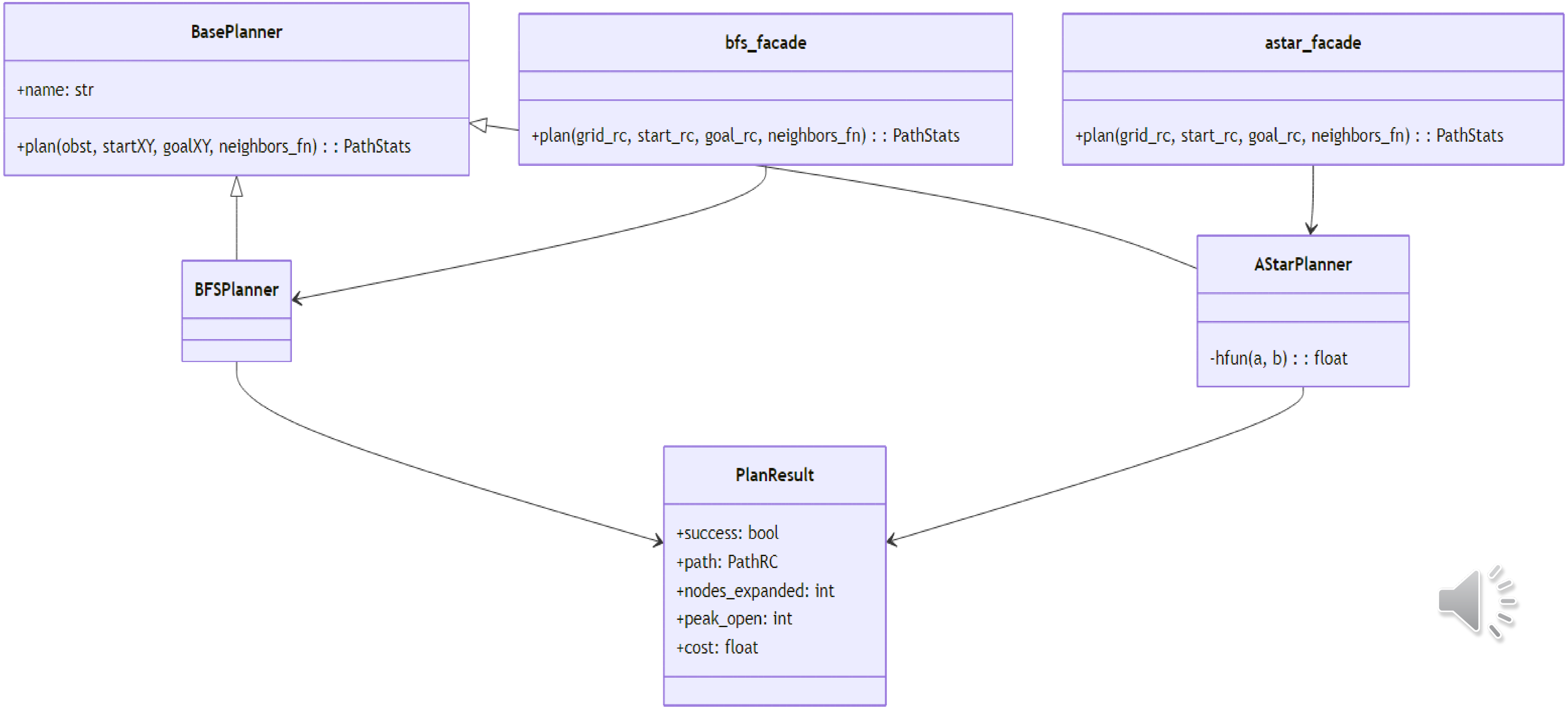
- Record metrics: **path length**, **runtime**, **nodes expanded**.

- Package the system using Docker for full reproducibility.

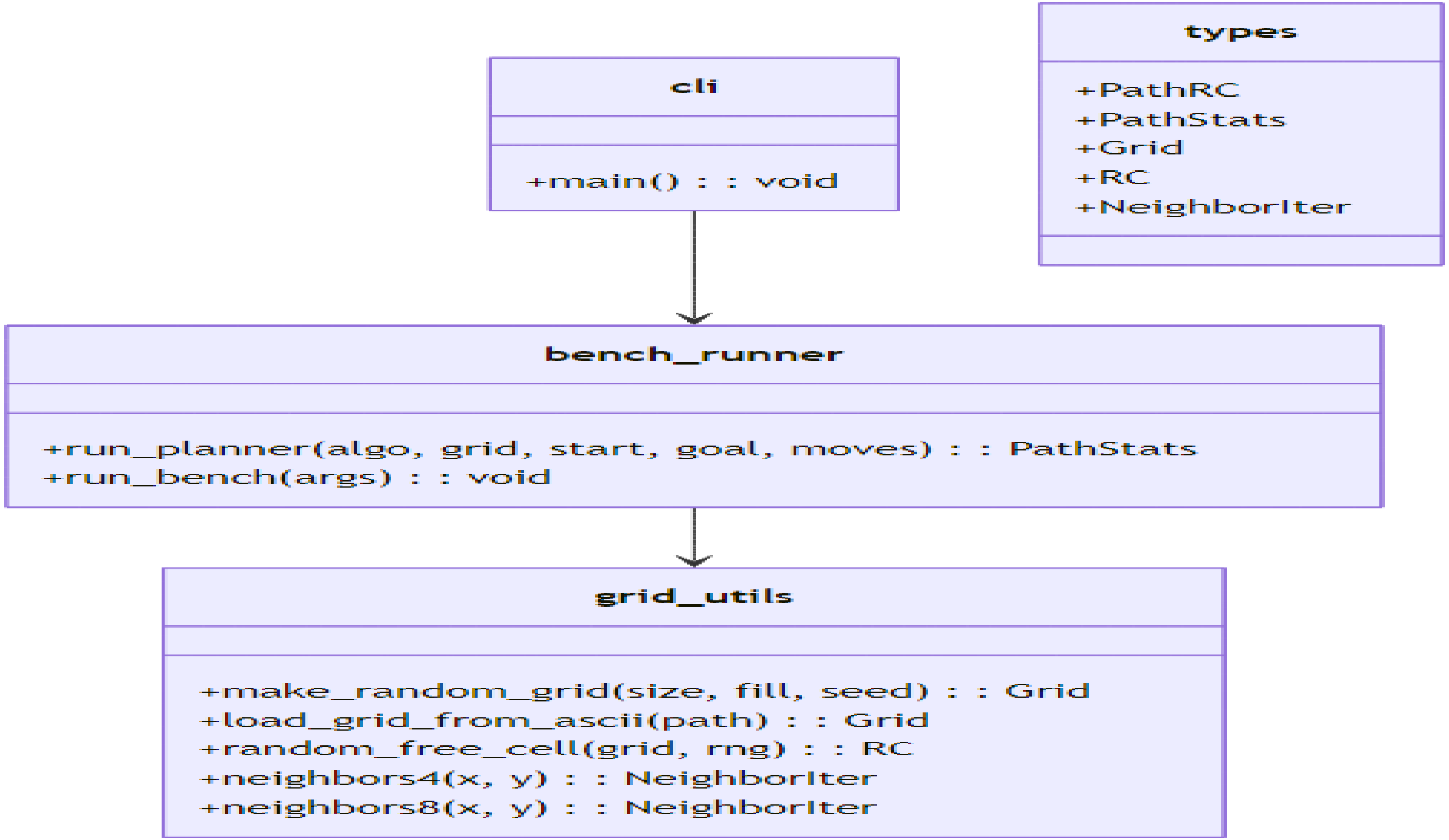
System Architecture



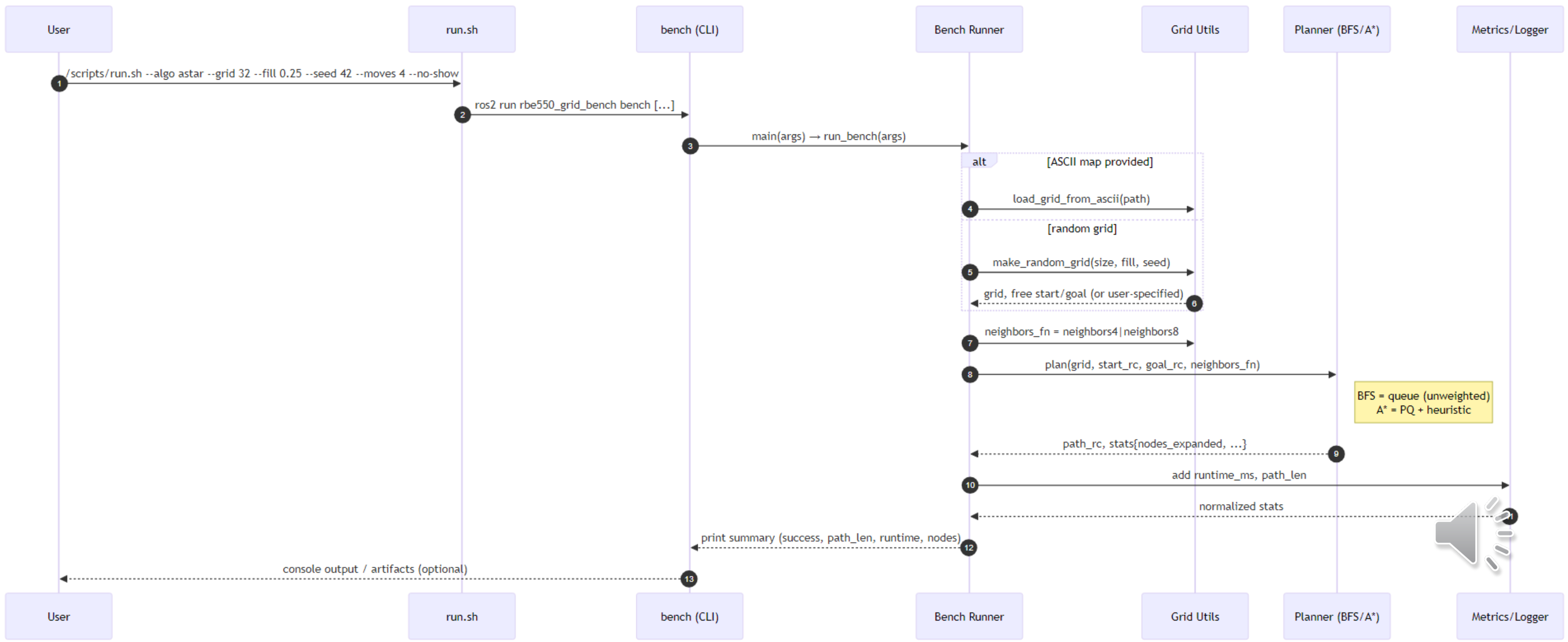
Class Structure (Algorithms)



Class Structure (Core & Environment)



Sequence Diagram



Packaging & Reproducibility

- **Entire benchmark is containerized for consistent runs across machines.**
- **A single command builds and executes all experiments.**
- **Docker image rbe550-bench includes:**
  - **ROS 2 Humble base**
  - **Workspace source (src/) and maps (maps/)**
  - **Automated build and execution via entrypoint**
- **Results are automatically stored in outputs/.**
- **Guarantees deterministic and fully reproducible benchmarking.**

Preliminary Results

Algorithm	Connectivity	Heuristic	Path Length	Nodes Expanded	Runtime (ms)
BFS	4	-	64	2155	2.39
A*	4	Manhattan	64	910	1.93
A*	8	Euclidean	45	455	1.81

- A* expands fewer nodes
- Heuristics reduce exploration
- 8-connected + Euclidean shortens path

Validation

- **Deterministic seeding for reproducible random maps.**
- **Maze and random environments tested repeatedly.**
- **Results are identical across different systems using Docker.**

Challenges & Next Steps

- **Challenges:**
- Fixing COLCON_TRACE build issue.
- Ensuring maps/ inclusion inside Docker.
- Balancing parameters for fair comparisons.

- **Next Steps:**
- Add weighted heuristics and cost variants.
- Integrate visualization (RViz / Matplotlib).
- Extend to probabilistic planners later in the semester.